

WON KO

kowon861@gmail.com

<https://won-ko.github.io/autonomous-driving-control-portfolio/>

SUMMARY

Senior Researcher focused on robotics and control, with experience in state estimation, stochastic optimal control, motion planning, and real-robot autonomy. Work spans EKF/ESKF, system identification, Frenet/B-spline trajectory tracking, and ROS 2 autonomous systems.

TECHNICAL SKILLS

Programming / Systems	C/C++, Python, MATLAB, OpenCV; ROS / ROS 2, Linux, Git, Docker, colcon
Control / Estimation	Stochastic Dynamic Programming, HJB/value iteration, system identification, EKF/ESKF, Kalman filtering, least squares, GTSAM/iSAM2
Planning / Prediction	Frenet/B-spline trajectory generation and tracking, obstacle avoidance, docking, vehicle-intent / trajectory prediction, classical ML / LSTM
Localization / SLAM	LIO-SAM, SC-LIO-SAM, LVI-SAM evaluation / LIO-VIO integration, GICP, Scan Context, intensity-assisted prior-map localization
Sensors / Calibration	Camera-LiDAR / RGB-D integration, IMU and LiDAR/base calibration, sensor synchronization, TF integration

EDUCATION

University of California, Santa Cruz Master of Science in Computer Engineering Emphasis in Robotics and Control M.S. Project: "Autonomously Driving Tricycle with Monocular Vision" Advisor: Prof. Dejan Milutinović	August 2022
University of California, Santa Cruz Bachelor of Science in Computer Engineering Concentration on Robotics and Control	December 2019

TEACHING EXPERIENCE

Teaching Assistant / University of California, Santa Cruz <i>Mechatronics (ECE 118)</i>	Spring 2022, Fall 2021 <i>Santa Cruz, CA</i>
- Led lab sections on event-driven programming, hierarchical state machines, sensor interfacing, and end-to-end system integration; provided debugging support and graded coursework with the TA team. - Maintained and repaired lab hardware; performed safety checks and managed spare parts/inventory to minimize downtime.	
Teaching Assistant / University of California, Santa Cruz <i>Computer Architecture (CSE 120)</i>	Spring 2021, Winter 2022 <i>Santa Cruz, CA</i>
- Led RISC-V and hardware-design labs using Verilog/SystemVerilog; guided students through simulation, debugging, and design verification. - Held weekly office hours and graded assignments/exams; supported performance analysis and correctness debugging for lab implementations.	

RESEARCH EXPERIENCE

Senior Researcher / Future Tech R&D Lab, Mobyus

November 2024 - Present

- Develop and validate autonomy software for AMR and autonomous-forklift platforms on real warehouse hardware, spanning state estimation, motion planning, obstacle avoidance, docking, perception, and system integration.
- Built and validated an ESKF-based LiDAR-IMU localization stack with ≤ 30 mm position error in controlled testbed evaluation; integrated navigation with Frenet/B-spline trajectory generation and tracking, pallet perception, and depth-based free-space estimation.
- Integrated trajectory tracking, obstacle avoidance, pallet docking, and PLC-facing execution; used calibration, trajectory comparison, timing/RMSE analysis, and failure diagnostics to validate behavior on physical robots.

Researcher / Robotics and Control Lab, UC Santa Cruz

September 2020 - August 2023

- Researched probabilistic state estimation and stochastic nonlinear control for GPS-free navigation under sensor noise and model uncertainty, including stochastic Dynamic Programming / HJB-style value iteration.
- Developed EKF and second-order nonlinear state estimators and performed system identification of a propeller-actuated mechanism using batch and recursive least squares in MATLAB/V-REP.

PROJECTS

Stochastic Optimal DP Controller for Autonomous Vehicle Tracking

Research / manuscript

- Formulated a stochastic Dynamic Programming controller with robot-frame relative-position / terminal-heading state, π turn-rate actions, diffusion noise, and a minimum-time/control-effort objective; the 34,596-state HJB policy converged in 1,156 iterations with 23/23 implementation tests passing.
- Across 2,400 matched simulations against two baselines, retained 100% terminal success and reduced mean arrival time by 2.2–4.9 s depending on the matched condition/baseline.

Autonomous Forklift Motion Planning & Real-Robot Validation

Mobyus, 2024–2025

- Integrated Frenet-frame / B-spline trajectory generation and tracking with LiDAR-IMU state estimation, obstacle avoidance, pallet docking, perception support, and PLC-facing execution on real warehouse hardware.

Robust Prior-Map Localization with Calibrated LiDAR Intensity

Ongoing R&D

- Built incident-angle/range-aware intensity calibration and keyframe-derived priormaps; used 20×60 Intensity Scan Context for bounded candidate/yaw prescoring before small-GICP geometric acceptance, with intensity MAE/RMSE diagnostics for failure analysis.

Camera-LiDAR Integration & Visual-Aided LIO / VIO

Ongoing Prototype

- Added synchronized RGB-D/depth-supported visual feature projection and keyframe persistence to the LIO pipeline, including FOV/depth association, timing, confidence, conflict, and occlusion diagnostics while keeping LiDAR-inertial odometry as the authoritative pose backbone.

Continental Intelligent Intersection – Vehicle Intent / Trajectory Prediction

Continental-sponsored UCSC

Project, 2018–2019

- Processed real intersection radar/camera data and applied classical ML and LSTM models to driver-intent / trajectory prediction; team-reported best LSTM accuracy was 89.5% before intersection entry and 98.6% after entry.

V2V Multi-Robot Collision Avoidance & Motion Coordination

Applied R&D

- Coordinated per-robot target speed using peer pose/path, conflict-region reasoning, ETA, priority, and safety margins; co-inventor, KR Application 10-2024-0041376 (pending/unexamined).